Assignment 4

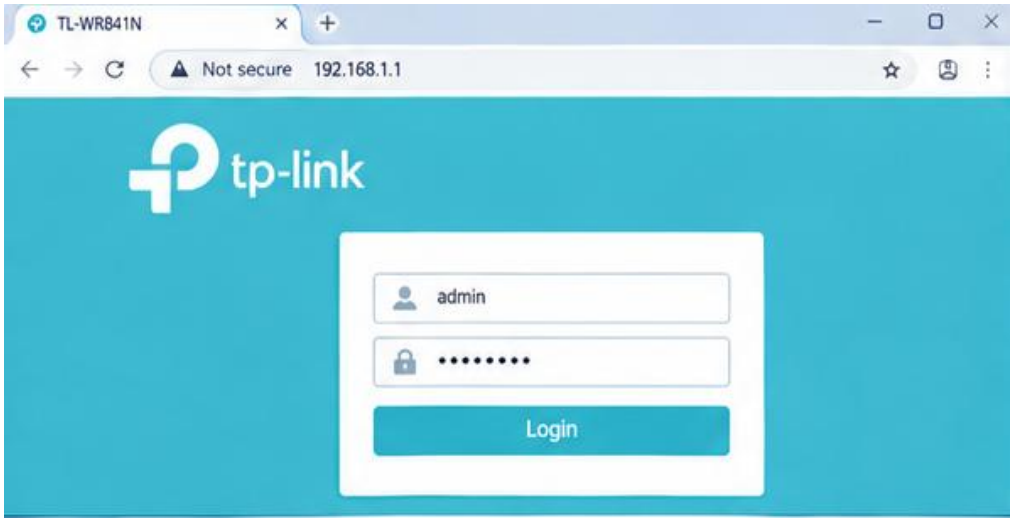
DAY 3: Configuring basic Network Hardware Theory (30 min): IP addressing basics and network configuration. Practical (30 min): Set up a router with default settings.

Task 1: Open Router Admin Panel

Objective:
To access the home Wi-Fi router's administration panel using its default IP address.

Steps:

1. Connect the laptop to your Wi-Fi network.
2. Press **Windows + R**.
3. Type cmd and press **Enter**.
4. Type:
   ipconfig
5. Find **Default Gateway**. Example: 192.168.1.1
6. Open Chrome/Edge.
7. Enter the Default Gateway IP in the address bar.
8. The router login page will appear.
9. Enter the router username and password if required.



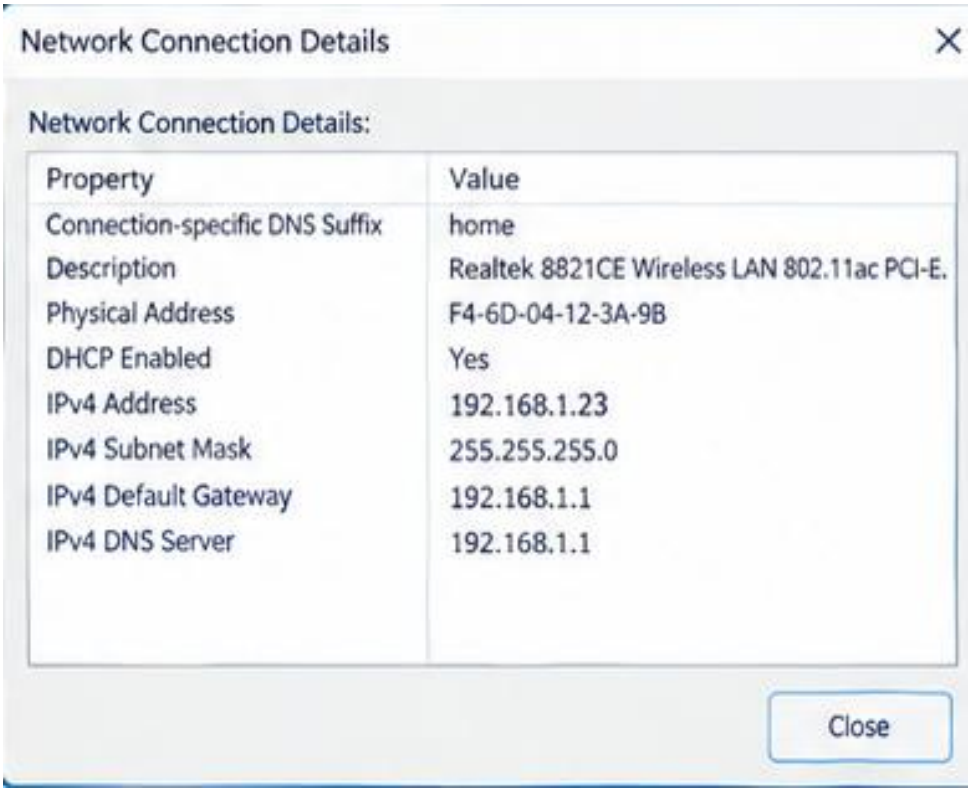
Task 2: Find IP Address, Subnet Mask and Default Gateway

Steps:

1. Open **Command Prompt**.
2. Type:
   ipconfig
3. Find the Wi-Fi adapter section.
4. Note these three values:
   - **IPv4 Address**
   - **Subnet Mask**
   - **Default Gateway**
   -

Example:

Network Information	Example Value
IPv4 Address	192.168.1.25
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1



Task 3: Change DHCP to Static IP

Objective:
To configure a static IP address and check whether internet access continues.

Example:
If router IP is 192.168.1.1, a suitable static IP could be:

- IP Address: 192.168.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- DNS: 8.8.8.8 *(example)*

Steps in Windows 11:

1. Open **Settings**.
2. Go to **Network & Internet → Wi-Fi**.
3. Select your connected Wi-Fi network.
4. Open **Hardware properties** / **IP assignment**.
5. Click **Edit** beside IP assignment.
6. Change **Automatic (DHCP)** to **Manual**.
7. Enable **IPv4**.
8. Enter the static IP details.
9. Save the settings.
10. Open a browser and check internet access.



Task 4: Check DHCP Range

Steps:

1. Open the router admin panel.
2. Go to **Network / LAN / DHCP Server** settings.
3. Find the **DHCP Server** section.
4. Note the **DHCP Start IP** and **DHCP End IP**.

Example:

DHCP Setting	Example
DHCP Start IP	192.168.1.100
DHCP End IP	192.168.1.200

